



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
08-11-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

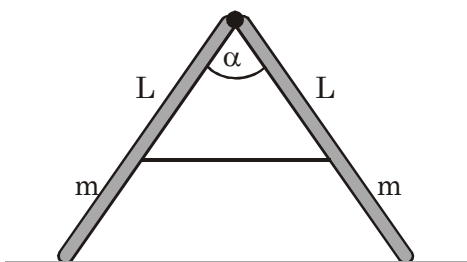
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

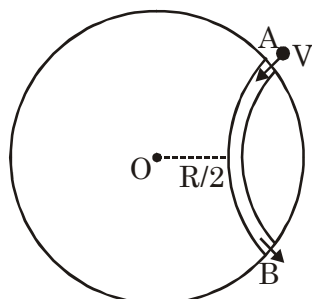
1. A step ladder is placed to the smooth horizontal surface of ice. The cross section of the ladder is the same every where and the angle between its legs is $\alpha = 60^\circ$. The hinge which connects the two legs of mass 5 kg each and of length of 2 m is frictionless. The thin, initially tight safety rope breaks and the ladder starts opening and moving downwards.



- (A) Just after breaking, the total normal force by ground is less than 100 N.
- (B) When the hinge does not exert any force on the legs, the angle between the legs of the ladder is 120°
- (C) When the hinge does not exert any force on the legs, the angle between the legs of the ladder is 180°
- (D) The hinge hits ice with a speed of $\sqrt{30\sqrt{3}}\text{ m/s}$.

Space for Rough Work

2. Assume that there is a tunnel in the shape of a circular arc through the earth. Wall of the tunnel is smooth. A ball of mass m is projected into the tunnel at A with minimum speed v such that the ball comes out of the tunnel at B and escapes out of the gravity of the earth. Mass and radius of the earth are M and R respectively and radius of the circle shaped tunnel is $\frac{3R}{4}$.



(A) $v = \sqrt{\frac{2GM}{R}}$

- (B) The normal force applied by the tunnel wall on the ball when it is closest to the centre of the earth is $\frac{27}{4} \frac{GMm}{R^2}$

- (C) At the initial moment, the speed is increasing at a rate of $\frac{GM}{R^2}$

- (D) At the initial moment, the force exerted by the wall is $\frac{8GMm}{3R^2}$

Space for Rough Work

3. A fixed motionless nonconducting thin-walled sphere of mass M is uniformly charged along surface with a full charge Q . Cut out of it a small disc whose mass and charge is $\frac{1}{n}$ times that of sphere, crush it into a tiny lump, place in the center of the sphere (the charge of the lump is saved) and released. Neglect gravity, radius of sphere is R ($n \gg 1$) :-

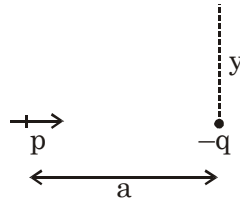
(A) The speed at a large distance from the sphere is approx $\sqrt{\frac{2kQ^2}{MRn}}$

(B) The initial acceleration of the lump is $\frac{kQ^2}{nMR^2}$.

(C) The acceleration of lump just out of the sphere is $\frac{kQ^2}{2MR^2}$.

(D) The speed of the lump goes on increasing

4. An electric dipole and point charge $-q$ are kept on x axis at a distance a from each other. Assume a is much larger than size of dipole. It is seen that $\vec{E}$ vanishes at a height y above $-q$:-



(A) $y = \sqrt{2} a$

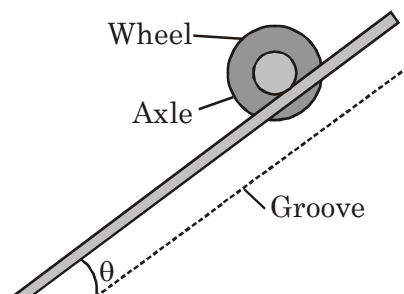
(B) $p = \frac{3\sqrt{3}}{2\sqrt{2}} qa$

(C) Potential energy of interaction of dipole with $-q$ is $\frac{-9\sqrt{3}k}{4a} q^2$

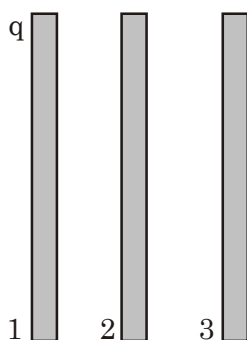
(D) Given situation is not possible.

Space for Rough Work

5. A uniform wheel of mass 10.0 kg and radius 0.400 m is mounted rigidly on an axle through its center (figure). The radius of the axle is 0.200 m and the rotational inertia of the wheel-axle combination about its central axis is 0.600 kg.m². The wheel is initially at rest at the top of a surface that is inclined at angle $\theta = 30.0^\circ$ with the horizontal; the axle rests on the surface while the wheel extends into a groove in the surface without touching the surface smoothly and without slipping. When the wheel axle combination has moved down the surface by 2.00 m.



- (A) Its rotational kinetic energy is 60 J. (B) Its translational kinetic energy is 30 J.
 (C) Its translational kinetic energy is 40 J. (D) Its rotational kinetic energy is 75 J.
6. Three metal plates are positively charged and laid parallel to each other as shown in figure. The net charge on the left plate is q . The net force on the middle plate is P . If the right plate was removed, the force acting on the middle plate will be equal to $\frac{P}{2}$. Find charges on each of the plates. The area of each plate is S . The field created by the plates is considered to be uniform.



- (A) $q_1 = q$ (B) $q_2 = \frac{PS\epsilon_0}{2q}$ (C) $q_3 = 3q$ (D) None of these

Space for Rough Work

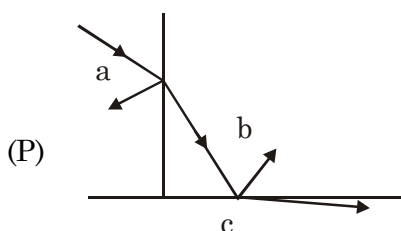
SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

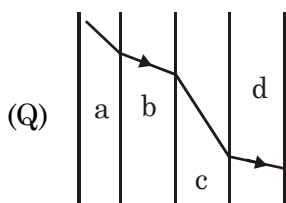
7. In each of situations below, the arrows indicate a light ray a, b and c are transparent medium of refractive index n_a , n_b & n_c respectively :-

List-I

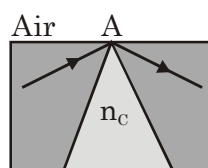
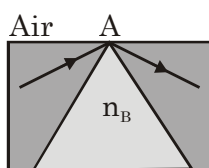
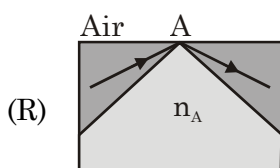
List-II



(1) $n_a > n_b > n_c$

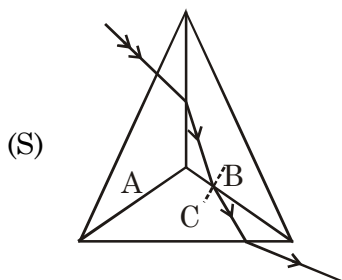


(2) $n_b > n_a > n_c$



(3) $n_b > n_c > n_a$

The dark region indicates the region for which light ray is totally reflected from the medium air boundary



(4) $n_a > n_c > n_b$

(5) $n_c > n_b > n_a$

(A) P → 1 ; Q → 2 ; R → 5 ; S → 1

(B) P → 1 ; Q → 2 ; R → 1 ; S → 1

(C) P → 1 ; Q → 3 ; R → 4 ; S → 5

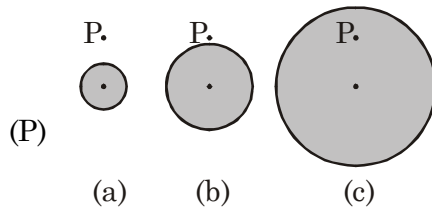
(D) P → 1 ; Q → 3 ; R → 4 ; S → 2

Space for Rough Work

8. In each of the options below, we have three charge configuration rank the magnitude of electric field at P in each case.

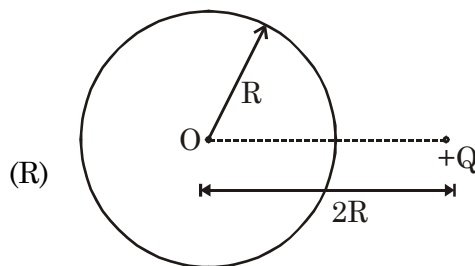
List-I

List-II



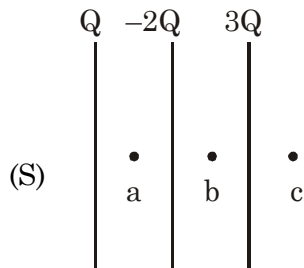
3 spheres with same charge Q uniformly distributed within the volume.

- (Q) An electric dipole at origin oriented along +ve x axis.
 $a \rightarrow (a, 0, 0)$, $b \rightarrow (-a, 0, 0)$; (c) $(a, a, 0)$



An uncharged spherical conductor centred at O as shown.

$a \rightarrow$ net electric field at O
 $b \rightarrow$ electric field at O due to $+Q$
 $c \rightarrow$ electric field at O due to induced charge.



3 large conducting plate carry charges as shown.

- (A) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 1$; $S \rightarrow 2$ (B) $P \rightarrow 3$; $Q \rightarrow 3$; $R \rightarrow 1$; $S \rightarrow 2$
 (C) $P \rightarrow 4$; $Q \rightarrow 4$; $R \rightarrow 1$; $S \rightarrow 1$ (D) $P \rightarrow 4$; $Q \rightarrow 4$; $R \rightarrow 3$; $S \rightarrow 4$

(1) $E_b = E_c > E_a$

(2) $E_b > E_c > E_a$

(3) $E_a = E_b > E_c$

(4) $E_a = E_b < E_c$

Space for Rough Work

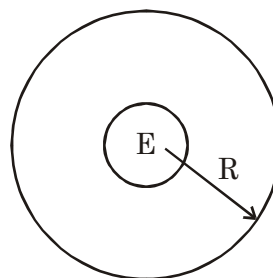
9. In column-II, certain paths of satellite around earth are shown. E_i represents the energy of satellite in i^{th} case, K_i represents the kinetic energy of satellite in i^{th} case, L_i represents its angular momentum and a_i represents semimajor axis.

List-I

List-II

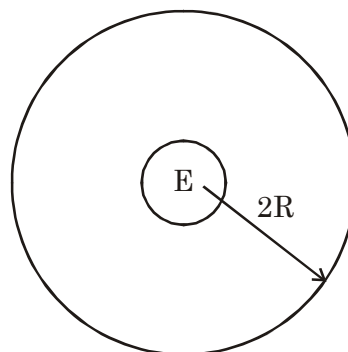
(P) a is maximum

(1)



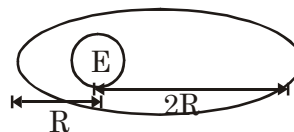
(Q) L is maximum

(2)



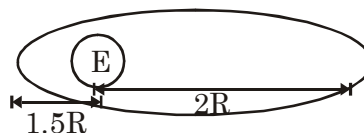
(R) K is maximum

(3)



(S) E is maximum

(4)



(A) $P \rightarrow 3$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 4$

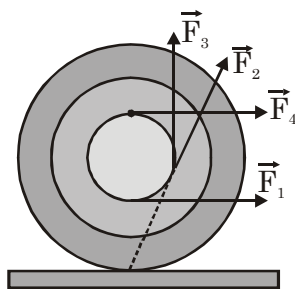
(B) $P \rightarrow 2$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 2$

(C) $P \rightarrow 2$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 1$

(D) $P \rightarrow 3$; $Q \rightarrow 1$; $R \rightarrow 4$; $S \rightarrow 2$

Space for Rough Work

10. A yo-yo rest on a rough horizontal surface. In each of the situation, assume that yo-yo does not slip. It can be subjected to 4 forces as shown. Consider only one force at a time.



List-I

(P) $\vec{F}_1$

(Q) $\vec{F}_2$

(R) $\vec{F}_3$

(S) $\vec{F}_4$

(A) $P \rightarrow 2,3$; $Q \rightarrow 3$; $R \rightarrow 1,3$; $S \rightarrow 2$

(C) $P \rightarrow 2,3$; $Q \rightarrow 3$; $R \rightarrow 1,4$; (S) $\rightarrow 1,4$

List-II

(1) yo-yo rolls to left

(2) yo-yo rolls to right

(3) Friction on yo-yo is to left

(4) Friction on yo-yo is to left

(B) $P \rightarrow 2,3$; $Q \rightarrow 3$; $R \rightarrow 1,3$; $S \rightarrow 1,3$

(D) $P \rightarrow 1,3$; $Q \rightarrow 2,3$; $R \rightarrow 1,4$; $S \rightarrow 4$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

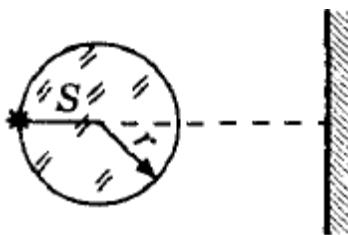
-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Diameter of a planet is $10d$, its mean density is $\frac{\rho_0}{4}$ and mass of its atmosphere is $10 m_0$, where d_0 , ρ_0 and m_0 are diameter, mean density and mass of atmosphere respectively for the earth. Assume that mean density of atmosphere is same on the planet and the earth and height of atmosphere on both the planets is very small compared to their radius. If a mercury barometer reads 76 cm on the surface of the earth, find its reading (in cm) on the surface of the planet.
2. An Astronomical unit is the distance between earth and sun assuming it to be a circular orbit. A comet is going around the sun in an elliptical orbit with a period of 64 year. The closest approach of the comet to the sun is 0.8 AU [AU = astronomical unit]. Calculate the greatest distance (in AU) of the comet from the sun.

Space for Rough Work

3. Inside the glass ball of radius $r = 2$ cm left of center near the surface is a point light source S (Fig). At what distance L from the center (in cm) to the right of the ball should you install a screen so that the radius of the illuminated region was equal to r ? Refractive index of the glass is 2.



4. On a glass bulb of an incandescent light bulb it is written: 220 V, 10 watts. Cold bulb filament resistance, measured with a multimeter turned out to be 484 ohms. What current (in A) will flow through this light bulb, if you connect it to a DC voltage of 110 V ? Assume that resistance is directly proportional to the absolute temperature of the filament and rate of heat loss is given by stefan's law.

Space for Rough Work

5. Very thin non conducting rod of length $0.75 R$ is uniformly charged along the length with charge of Q . A small conductive ring of radius R is made of very thin wire, its center coincides with one of the ends of the rod, and the plane of the ring is perpendicular to the rod. Ring carries charge q .
What force (in N) does the rod exert on ring? (Take : $\frac{kQq}{R^2} = 8N$)
6. A light bulb with power 100 W is designed for an input voltage of 110 V . What is the resistance of the resistor (in Ω) that should be connected in series to the light bulb such that it has the same brightness with an input voltage of 127 V ?

Space for Rough Work

7. A plano-convex lens made of glass has a refractive index $n = 1.5$ and a diameter $D = 5$ cm. The radius of the convex spherical surface $R = 5$ cm. A broad parallel beam of light hits the flat side of the lens along its optic axis. Calculate the diameter (in cm) of the light spot formed on a screen set behind the lens perpendicular to the incident beam. The screen is positioned so as to obtain the smallest light spot for a narrow beam (restricted by a diaphragm) directed along the optic axis.
8. A large, thin conducting plate with area S and thickness d is placed in a homogeneous electrical field E , which is perpendicular to the plate. How much heat is dissipated in the plate when the field is switched off? If answer is $k\epsilon_0 E^2 S d$ fill k in OMR sheet.

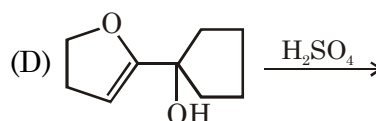
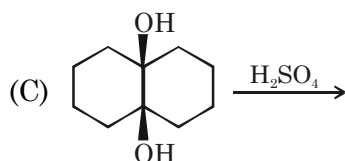
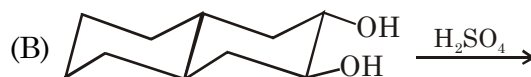
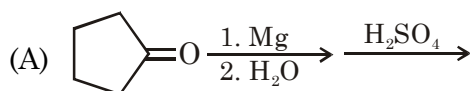
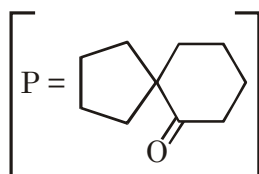
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following reaction sequence gives (P) as a final major product :

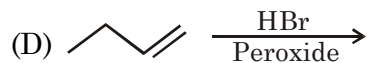
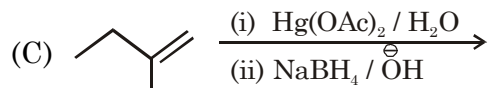
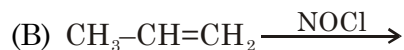
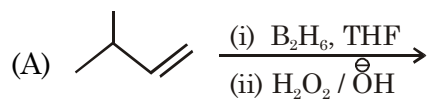


Space for Rough Work

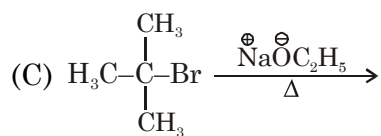
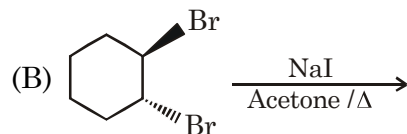
2. Choose the **INCORRECT** statement(s).
- (A) The shapes of PCl_4^+ , PCl_4^- and AsCl_5 are respectively tetrahedral, square bipyramidal and trigonal bipyramidal
- (B) ICl_2^- has three equatorial lone pairs and two axial bond pairs on central atom
- (C) CO_2 , N_3^- and NO_2^+ are linear but I_3^- is not
- (D) The smallest bond angle of CH_2F_2 is smaller than smallest bond angle of CHF_3
3. Correct statement(s) out of following about face centred cubic unit cell of an element of edge length a (Take $\sqrt{2} = 1.414$, $\sqrt{6} = 2.45$)
- (A) Packing fraction is 0.74
- (B) Fraction of edge that remain unoccupied is approximately 0.293
- (C) Distance of third nearest neighbour of an atom is $\sqrt{1.5}a$
- (D) Difference in distance between two nearest tetrahedral void and two nearest octahedral voids is $0.207a$.
4. 2 moles of liquid A ($P_A^0 = 100$ torr) and 3 moles of liquid B ($P_B^0 = 300$ torr) are mixed to form an ideal solution. The vapour pressure of distillate of above ideal solution is -
- (A) 263.64 torr (B) 220 torr (C) 180 torr (D) 136.36 torr

Space for Rough Work

5. In which of the given reactions final product is corresponding to anti Markonikoff's rule :



6. Which of the following reaction pretominately follow E_2 mechanism :



Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match the list:

List- I	List- II
(P) Hypophosphorus acid	(1) Basicity of acid ≥ 3
(Q) Trimetaphosphoric acid	(2) P–O–P linkage is present
(R) Peroxydiphosphoric acid	(3) P–P linkage is present
(S) Tetrapolyphosphoric acid	(4) Atleast one P–H bond is present
	(5) Atleast one P-atom has +5 oxidation state

The correct option is

- (A) P $\rightarrow$ 5; Q $\rightarrow$ 1,2,5; R $\rightarrow$ 1,5,6; S $\rightarrow$ 1,3
(B) P $\rightarrow$ 4,5; Q $\rightarrow$ 1,2,3; R $\rightarrow$ 1,5; S $\rightarrow$ 1,2,5
(C) P $\rightarrow$ 4; Q $\rightarrow$ 1,2; R $\rightarrow$ 1,5; S $\rightarrow$ 1,3
(D) P $\rightarrow$ 4; Q $\rightarrow$ 1,2,5; R $\rightarrow$ 1,5; S $\rightarrow$ 1,2,5

Space for Rough Work

8. List-I

- (P) Electron moving in 2nd orbit in He⁺ ion
- (Q) Electron moving in 3rd orbit in H-atom
- (R) Electron moving in 1st orbit in Li⁺² ion
- (S) Electron moving in 2nd orbit in Be⁺³ ion

List-II

- (1) Radius of orbit in which electron is moving is 0.529 Å
- (2) Total energy of electron is $-13.6 \times 9\text{eV}$
- (3) Velocity of electron is $\frac{2.188 \times 10^6}{3} \text{m/sec}$
- (4) De-broglie wavelength of electron is $\sqrt{\frac{150}{13.6}} \text{Å}$

The correct option is

- (A) P → 4, Q → 3, R → 2, S → 1
- (B) P → 3, Q → 4, R → 2, S → 1
- (C) P → 4, Q → 3, R → 1, S → 2
- (D) P → 1, Q → 2, R → 3, S → 4

Space for Rough Work

9. List-I

- (P) Na_2O
 (Q) CaF_2
 (R) ZnS
 (S) NaCl

The correct option is

- (A) $P \rightarrow 4;$ $Q \rightarrow 3;$ $R \rightarrow 1;$ $S \rightarrow 2$
 (B) $P \rightarrow 2;$ $Q \rightarrow 4;$ $R \rightarrow 1;$ $S \rightarrow 3$
 (C) $P \rightarrow 3;$ $Q \rightarrow 4;$ $R \rightarrow 1;$ $S \rightarrow 2$
 (D) $P \rightarrow 2;$ $Q \rightarrow 3;$ $R \rightarrow 1;$ $S \rightarrow 4$

List-II (TV = tetrahedral voids)

(OV = octahedral voids)

- (1) Anions = FCC, Cations = Alternate 'TV's
 (2) Anions = FCC, Cation = All 'OV's
 (3) Cations = FCC, anion = all 'TV's
 (4) Anions = FCC, Cation = All 'TV's

Space for Rough Work

10.	List-I (Reactions)	List-II (Favourable conditions)
	(P) Oxidation of nitrogen $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) + 180.5 \text{ kJ} \rightleftharpoons 2\text{NO}(\text{g})$	(1) Addition of inert gas at constant pressure
	(Q) Dissociation of $\text{N}_2\text{O}_4(\text{g})$ $\text{N}_2\text{O}_4(\text{g}) + 57.2 \text{ kJ} \rightleftharpoons 2\text{NO}_2(\text{g})$	(2) Decrease in pressure
	(R) Oxidation of $\text{NH}_3(\text{g})$ $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g}) + 905.6 \text{ kJ}$	(3) Decrease in temperature
	(S) Formation of $\text{NO}_2(\text{g})$ $\text{NO}(\text{g}) + \text{O}_3(\text{g}) \rightleftharpoons \text{NO}_2(\text{g}) + \text{O}_2(\text{g}) + 200 \text{ kJ}$	(4) Increase in temperature

Code :

- | | | | |
|--------------|------------|------------|-------|
| (A) P → 1; | Q → 2,4; | R → ,2,3; | S → 4 |
| (B) P → 3,4; | Q → 1,2; | R → 1,2,3; | S → 4 |
| (C) P → 4; | Q → 1,2,4; | R → 1,2,3; | S → 3 |
| (D) P → 2; | Q → 1,2,4; | R → 1,2,3; | S → 3 |

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Three elements P, Q and R have atomic numbers Z_1 , Z_2 and Z_3 respectively.

If $Z_1 - Z_2 = 2$ and $\frac{Z_1 + Z_2}{2} = Z_3 - 2$

and the electronic configuration of element P is $[\text{Ar}]3d^8 4s^2$ then

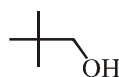
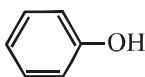
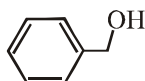
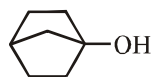
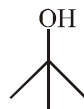
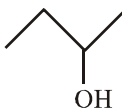
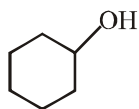
t = maximum number of electrons present in element P for which $\ell = 1$

s = number of paramagnetic species among P, Q and R

Find the value of (t + s)

Space for Rough Work

2. How many alcohols among following gives turbidity with Lucas reagent in less than 10 minutes :



3. In open container of volume 100 ml, air is present at 300K. If container is heated to 400 K then calculate volume of air (in ml) escape out if it is measured at 5 atm pressure and 600 K temperature.
4. Consider the following

(A) CCl_3^- , NH_3 , $\text{CO}_2 \Rightarrow$ P = Number of pure atomic orbital(s) of central atom used in bonding for non polar specie.

(B) F, Ne, Na $\Rightarrow$ Q = Atomic number of specie having smallest IE

(C) HOI , HOBr , $\text{HOCl} \Rightarrow$ $R = \frac{z}{2}$ where z is the atomic number of halogen corresponding to weakest acid

then calculate the value of $(R-Q+P)$.

Space for Rough Work

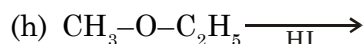
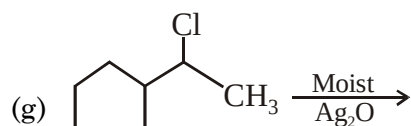
5. How many compounds decolorise Br_2 water solution ?

- (a) Pent-2-yne (b) Hexa-2,4-diene (c) Cyclohexene (d) Styrene $\left(\text{C}_6\text{H}_5\text{CH=CH}_2 \right)$
- (e) Benzene (f) n-pentane (g) Trans-But-2-ene (h) Ethane
- (i) Cyclopentane

6. 1 mole of H-atoms [atom is in ground state] are excited by bombarding photons of energy 8.4 eV, 12.09 eV and 15 eV of energy, then the number of different wave lengths emitted is equal to.

7. How many reactions giving alkyl halide as one of the final product ?

- (a) $\text{C}_2\text{H}_5\text{OH} + \text{SOCl}_2 \longrightarrow$ (b) $\text{R-I} + \text{AgF} \longrightarrow$
- (c) $\text{EtOH} + \text{NaI} + \text{H}_2\text{SO}_4 \longrightarrow$ (d) $\text{EtOH} + \text{NaI} + \text{H}_3\text{PO}_4 \longrightarrow$
- (e) $\text{CH}_3\text{-CH=CH}_2 \xrightarrow{\text{HBr}}$ (f) $\text{CH}_3\text{-CH}_2\text{-Cl} \xrightarrow[\text{(Acetone)}]{\text{NaI}}$



8. A^{n+} is oxidised by acidified KMnO_4 solution into AO_3^- . If 2.68 milli moles A^{n+} requires 32.16 mL of a 0.05 M acidified KMnO_4 solution for complete oxidation, find the value of n.

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For the equation $\sin x + \cos x = -\frac{1}{3}$, which of the following holds good ?

(A) The possible value of $\tan x$ is $\frac{\sqrt{17} + 1}{1 - \sqrt{17}}$

(B) The possible value of $\tan x$ is $\frac{1 - \sqrt{17}}{\sqrt{17} + 1}$

(C) There exists exactly two solutions of the equation in $[0, 2\pi]$.

(D) There exists exactly four solutions of the equation in $[0, 2\pi]$.

Space for Rough Work

2. Let $f(x) = \begin{cases} \min\{\sin^{-1} x, \cos^{-1} x\} & ; \quad 0 \leq x \leq 1 \\ \max\{\sin^{-1} x, \cos^{-1} x\} & ; \quad -1 \leq x < 0 \end{cases}$ and $f(x) = \frac{1}{2}$, $f(x) = \frac{1}{4}$ & $f(x) = 2$ has number of solution(s) $\lambda_1, \lambda_2, \lambda_3$ respectively, then $100\lambda_1 + 200\lambda_2 + 300\lambda_3$ is less than -
 (A) 1200 (B) 200 (C) 600 (D) 1000
3. Let $S_n = \sum_{r=2}^n \frac{3^{r-1}(2r-3)}{r(r-1)}$, then
 (A) S_9 is divisible by 4 (B) S_9 is divisible by 21
 (C) $10.S_{10} + 3 = 3^{10}$ (D) $7(S_7 + 3)$, $10(S_{10} + 3)$, $13(S_{13} + 3)$ are in G.P.
4. Which of the following statement(s) is (are) correct ?
 (A) $\cos^{-1}(4x^3 - 3x) = 3\cos^{-1}x \Rightarrow x \in \left[\frac{1}{2}, 1\right]$
 (B) If $g(x)$ is periodic function, then $f(g(x))$ is periodic function.
 (C) If $f(x) = x^3 - x^2 + x + 1$, then $f^{-1}(2)$ is 1
 (D) The equation $f(x) = f^{-1}(x)$ and $f(x) = x$ always gives same set of solutions ('f' is bijective function)

Space for Rough Work

5. Sum of the series $({}^{100}C_1)^2 + 2.({}^{100}C_2)^2 + 3.({}^{100}C_3)^2 + \dots + 100.({}^{100}C_{100})^2$ equals-
- (A) $\frac{2^{99}[1.3.5.....(199)]}{99!}$ (B) $100.{}^{200}C_{100}$
- (C) $50.{}^{200}C_{100}$ (D) $100.{}^{199}C_{99}$
6. Equation of a minor axis of an ellipse is $x + 2y = 3$ and length of minor axis is $\sqrt{5}$, foot of directrix is $(2, 3)$, then
- (A) equation of ellipse is $4(2x - y - 1)^2 + 2(x + 2y - 3)^2 = 25$
- (B) eccentricity of ellipse is $\frac{1}{\sqrt{2}}$
- (C) major axis is $\sqrt{10}$
- (D) length of LR is $\frac{\sqrt{5}}{\sqrt{2}}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Consider a ΔABC , with usual notation.

List-I

List-II

- | | |
|---|--|
| (P) $b = 2, c = \sqrt{3}, \angle A = \frac{\pi}{6}$, then value of R is | (1) 9 |
| (Q) $A = \frac{2\pi}{3}, b - c = 3\sqrt{3}, \Delta = \frac{9\sqrt{3}}{2}$, then a is | (2) 1 |
| (R) $\Delta = 100, r_1 = 10, r_2 = 50$, then $(b - a)$ is | (3) 2 |
| (S) line joining the circumcenter and incenter parallel to BC, then $\cos B + \cos C$ is equal to | (4) 8 |
| | (5) 3 |
| (A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$ | (B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 4$ |
| (C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 5$ | (D) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$ |

Space for Rough Work

8. For the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 16x - 16y + 127 = 0$.

List-I

List-II

(P) Length of direct common tangent is less than

(1) $\sqrt{128}$

(Q) Length of transverse common tangent is

(2) $\sqrt{119}$

(R) Point of intersection of direct common tangents is

(3) (16, 16)

(S) Point of intersection of transverse common tangent is

(4) $\left(\frac{16}{3}, \frac{16}{3}\right)$

(5) $\sqrt{135}$

(A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 1, 5; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

Space for Rough Work

9. Normals of parabola $y^2 = 4x$ at P and Q meet at R ($x_2, 0$) and tangents at P and Q meet at T ($x_1, 0$) if $x_2 = 3$, then

Match the following:

List-I

List-II

- | | |
|--|-------|
| (P) The area of quadrilateral PTQR is | (1) 3 |
| (Q) Value of $[PT]$ where $[.]$ greatest integer function | (2) 2 |
| (R) If quadrilateral PTQR can be inscribed in a circle, then the value of (circumference of circle) / 4π will be | (3) 1 |
| (S) The number of normals that can be drawn to the parabola from R($x_2, 0$) is | (4) 8 |
| (A) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$ | (5) 4 |
| (B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$ | |
| (C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$ | |
| (D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$ | |

Space for Rough Work

10. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

(P) $f(x) = x \sin x$

(Q) $f(x) = x + \tan^{-1}x$

(R) $f(x) = \{\cos x\}$

(S) $f(x) = \{\tan x\}$

(Note : $\{.\}$ denotes fractional part function)

List-II

(1) Even

(2) Bounded

(3) Periodic

(4) Invertible on $\mathbb{R} \rightarrow \mathbb{R}$

(5) Odd

(A) $P \rightarrow 1,2; Q \rightarrow 4,5; R \rightarrow 1,2; S \rightarrow 2,3$ (B) $P \rightarrow 1; Q \rightarrow 4,5; R \rightarrow 1,2,3; S \rightarrow 2,3$

(C) $P \rightarrow 1,2; Q \rightarrow 5; R \rightarrow 1,2; S \rightarrow 3$ (D) $P \rightarrow 1; Q \rightarrow 5; R \rightarrow 1,2,3; S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

										+	●
●	●	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

										●	-
●	●	●	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- The whole domain of $y = f(x)$ defined implicitly by the equation $3^y + 2^{x^4} = 2^{4x^2-1}$ is $(a,b) \cup (c,d)$, then $a + b + c + d$ is
- Let $f(x)$ be an odd function defined on $\mathbb{R}$ such that $f(x) = x^2$ when $x \geq 0$. If $f(x+a) \geq 2f(x)$ holds $\forall x \in [a, a+2]$, then the least integral value of a is

Space for Rough Work

3. If α, β where $\alpha > \beta$ are roots of $x^2 - 4x + 2 = 0$, then value of determinant

$$\begin{vmatrix} 1 + \alpha - \beta & 2\sqrt{\alpha\beta} & -2\sqrt{\beta} \\ 2\sqrt{\alpha\beta} & 1 - \alpha + \beta & 2\sqrt{\alpha} \\ 2\sqrt{\beta} & -2\sqrt{\alpha} & 1 - \alpha - \beta \end{vmatrix} \text{ is}$$

4. The equation $\sin \cot^{-1} \cos \tan^{-1} x = a$ has real solution iff complete range of a is $[p, q)$. Value of $\left[q + \frac{1}{p} \right]$ is (where $[.]$ denotes greatest integer function)

5. If $\sin^{-1} \frac{1}{\sqrt{10}} + \sin^{-1} \frac{2}{\sqrt{85}} + \sin^{-1} \left(\frac{4}{\sqrt{1105}} \right) + \dots \dots \dots \infty = \alpha$, then the value of $\tan \alpha$ is

Space for Rough Work

6. Consider $(1 - y)^m(1 + y)^n = 1 + a_1y + a_2y^2 + \dots$ where $m, n \in \mathbb{N}$ and $a_1 = a_2 = 10$ then $\left(\frac{m+n}{10}\right)$ is equal to
7. The coefficient of x^{401} in $(1 + x + x^2 + \dots + x^9)^{-1}$ is
8. Let number x & $x + 32$ are three digit & four digit palindromes respectively then the sum of the digits of x is equal to
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Space for Rough Work

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QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two section as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator